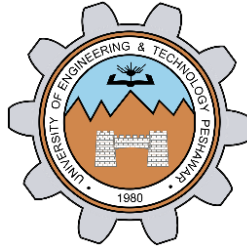


4 digit BCD Counter on Multiplexed Seven Segment Display

LAB # 08



Spring 2024

CSE-308L Digital System Design Lab

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Registration No.: **21PWCSE2059**

Class Section: **C**

“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

Dr. Asif Ali Khan

Date:

5th May 2024

Department of Computer Systems Engineering
University of Engineering and Technology, Peshawar

Objectives:

This lab will enable students to:

- Learn to use time multiplexed 4 digit Seven Segment display

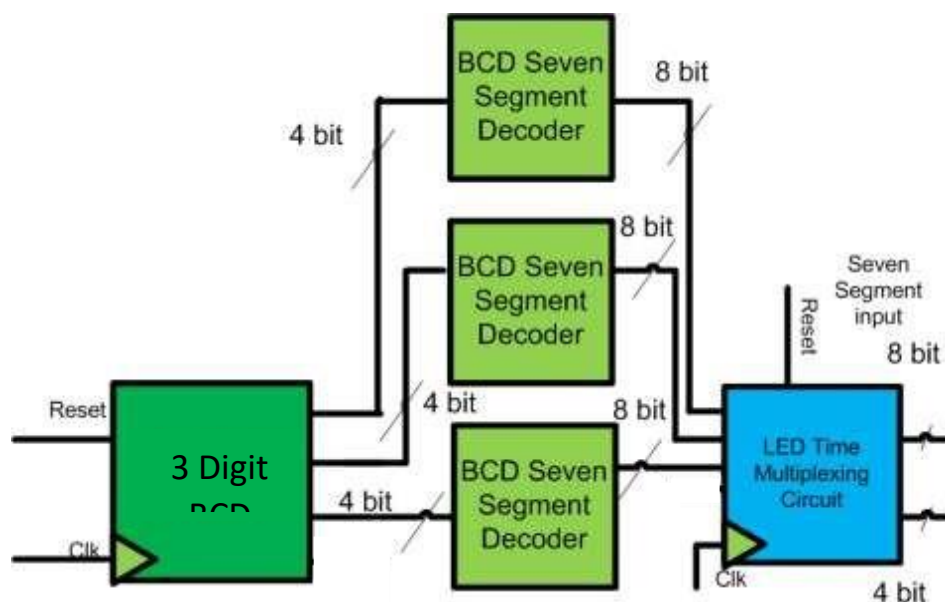
Software used:

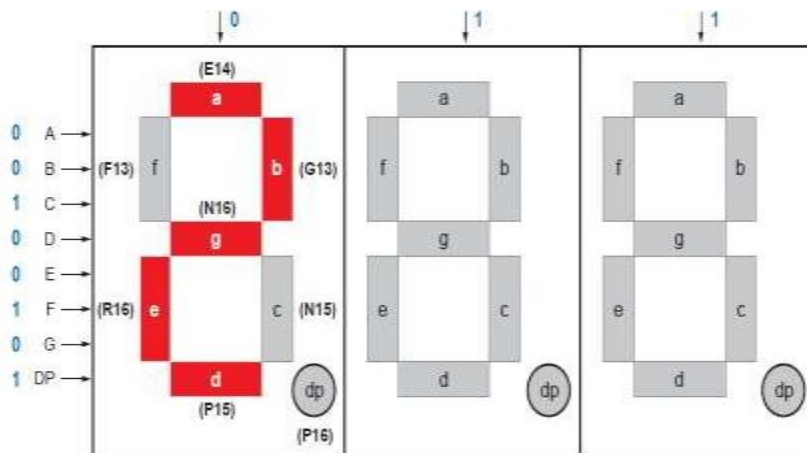
- Xilinx ISE

Block Diagram:

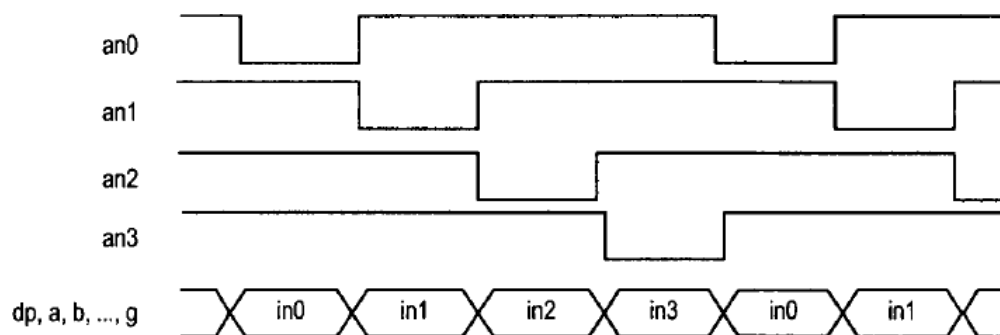
The S6 board has four seven-segment LED displays, each containing seven bars and one small round dot. To reduce the use of FPGA's I/O pins, the S6 board uses a time-multiplexing sharing scheme. In this scheme, the three displays have their individual enable signals but share eight common signals to light the segments. All signals are active low (i.e., enabled when a signal is 0). The schematic of displaying a "2" on the leftmost LED is shown in Figure.

Note that the enable signal (i.e., an) is "011". This configuration clearly can enable only one display at a time. We can time-multiplex the four LED patterns by enabling the four displays in turn, as shown in the simplified timing diagram. If the refreshing rate of the enable signal is fast enough, the human eye cannot distinguish the on and off intervals of the LEDs and perceives that all four displays are lit simultaneously. This scheme reduces the number of I/O pins from 32 to 11 (i.e., eight LED segments plus three enable signals) but requires a time-multiplexing circuit.





Timing Diagram of LED Multiplexing:



LED time Multiplexing:

The refresh rate of the enable signal has to be fast enough to fool our eyes but should be slow enough so that the LEDs can be turned on and off completely. The rate around the range 1000 Hz should work properly. In our design, we use an 18-bit binary counter for this purpose. The two MSBs are decoded to generate the enable signal and are used as the selection signal for multiplexing

Lab Task:

Implement a BCD counter that runs from 000 to 999 and shows each BCD digit on the seven segment display.

Code:

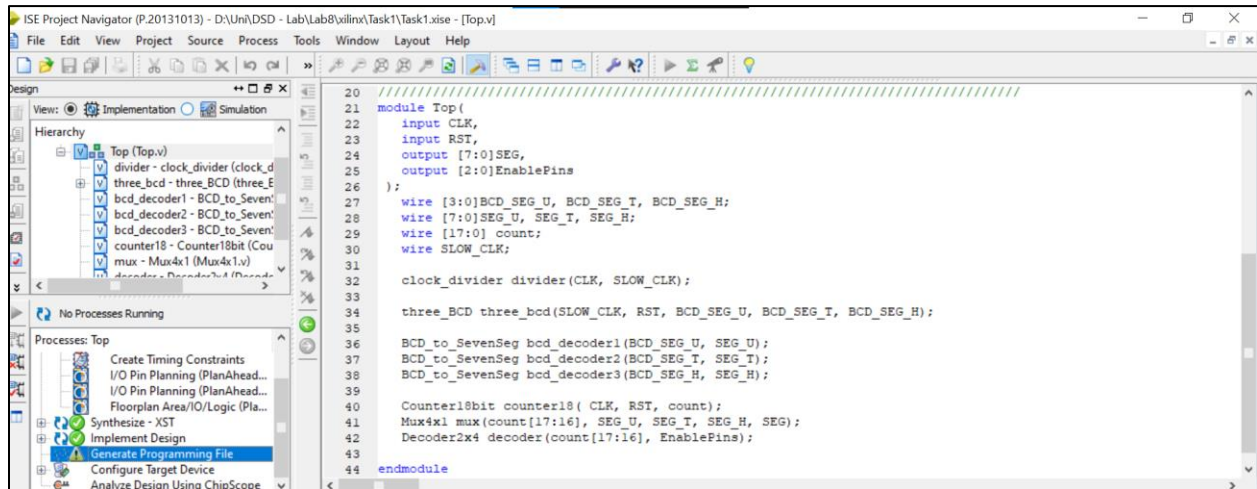


Figure 1.1 top module

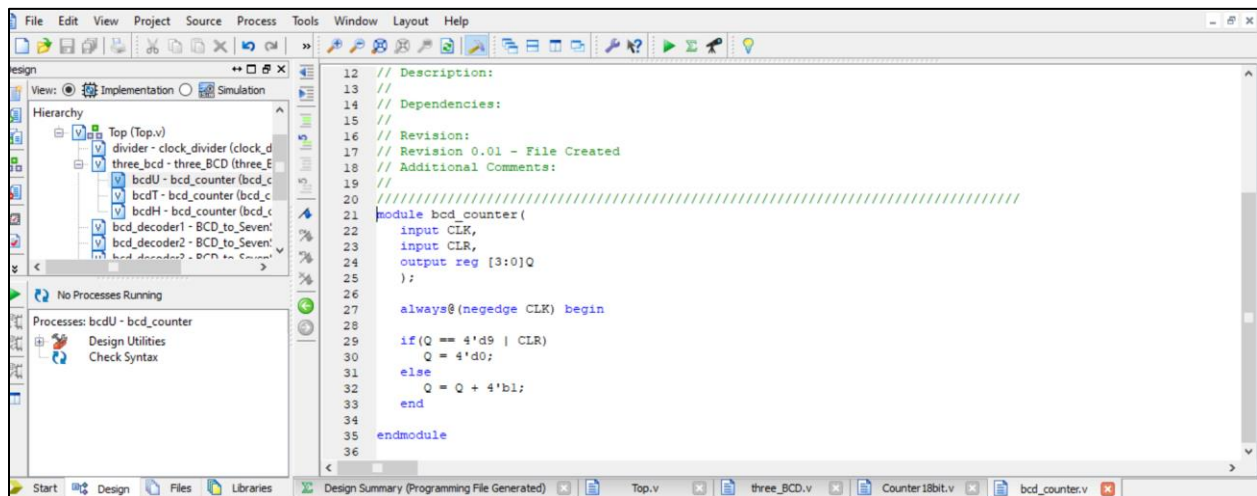


Figure 1.2: BCD Counter Code

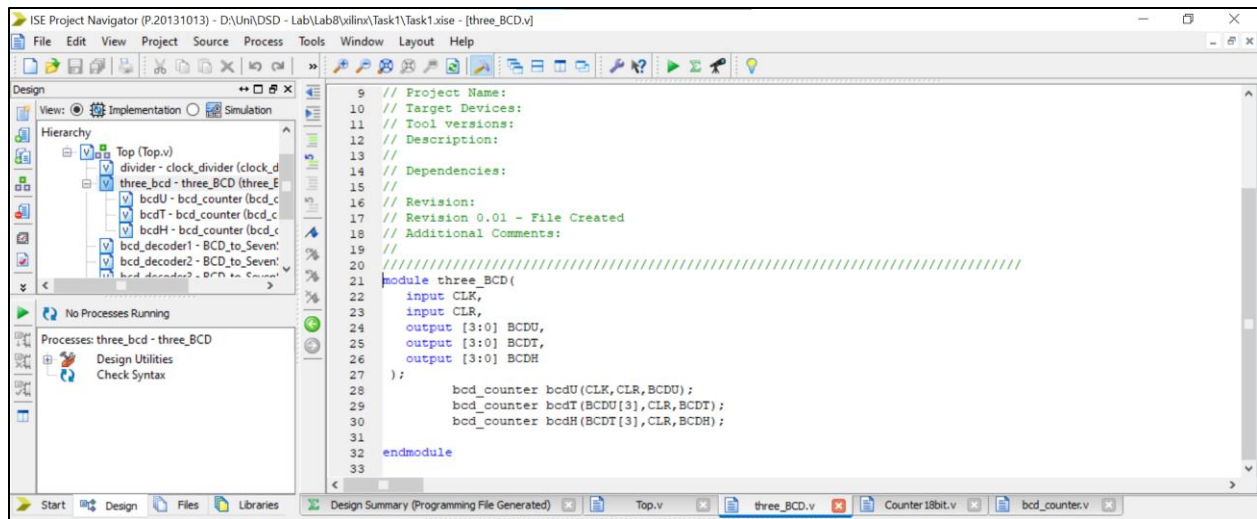


Figure 1.3: Three BCD Counter Code

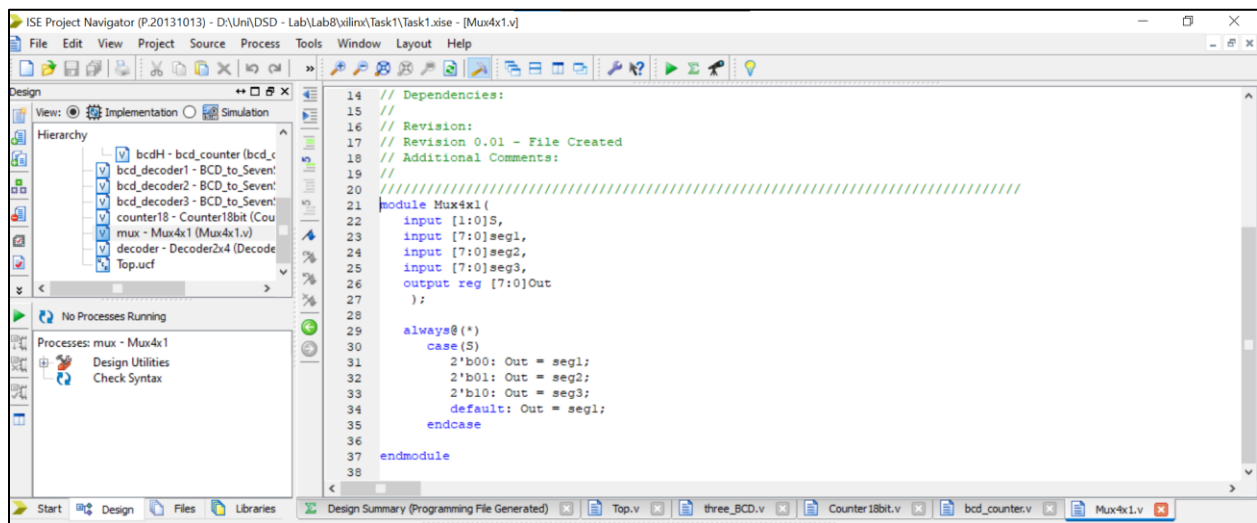


Figure 1.4: 4x1 Multiplexer Code

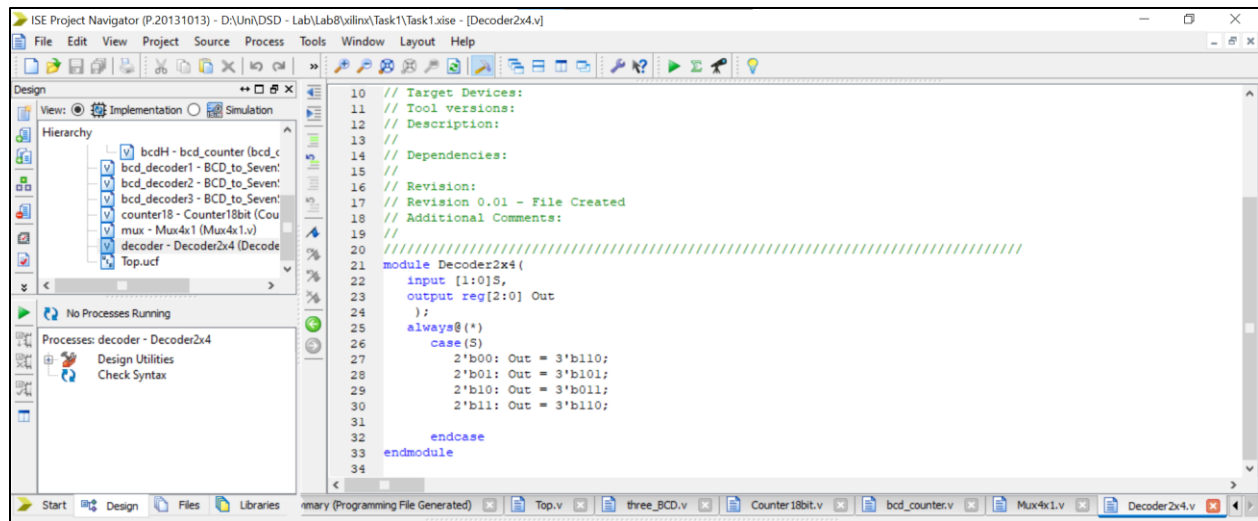


Figure 1.5: 2x4 Decoder Code

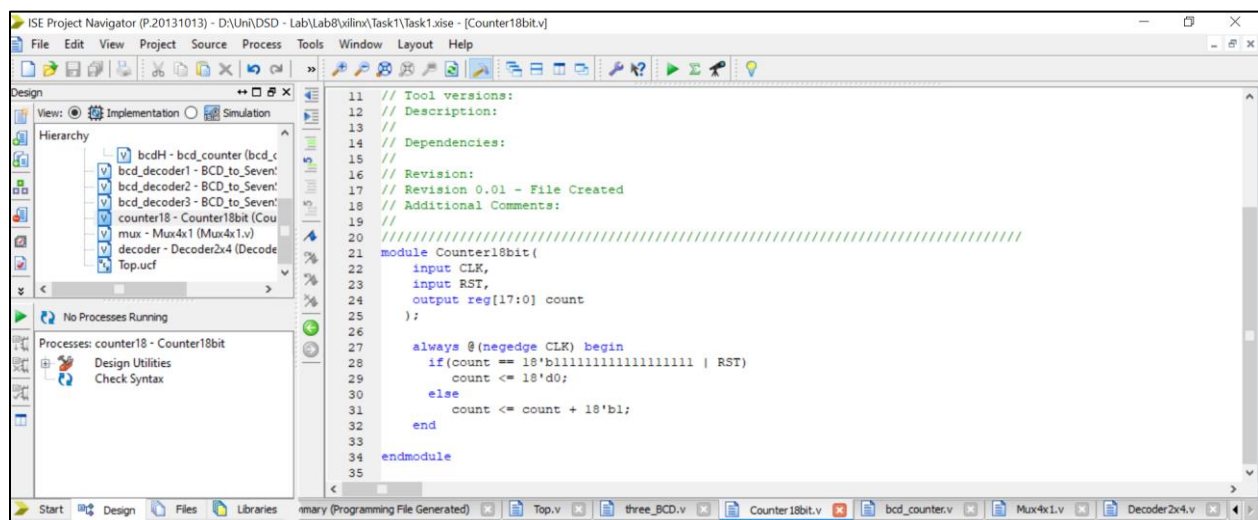


Figure 1.6: 18 Bit Counter Code

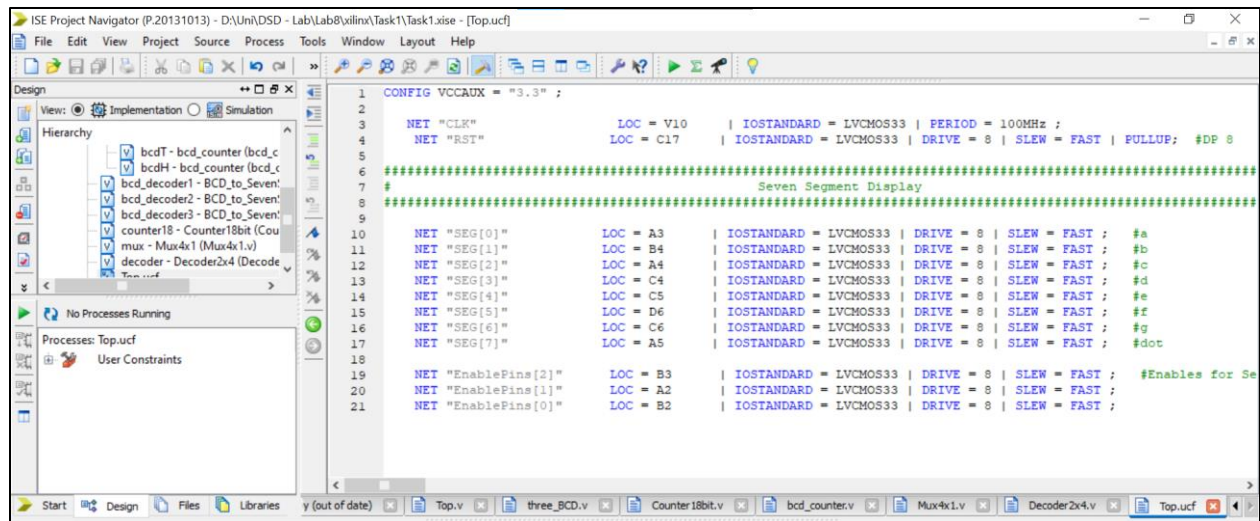


Figure 1.7: UCF File

Output:

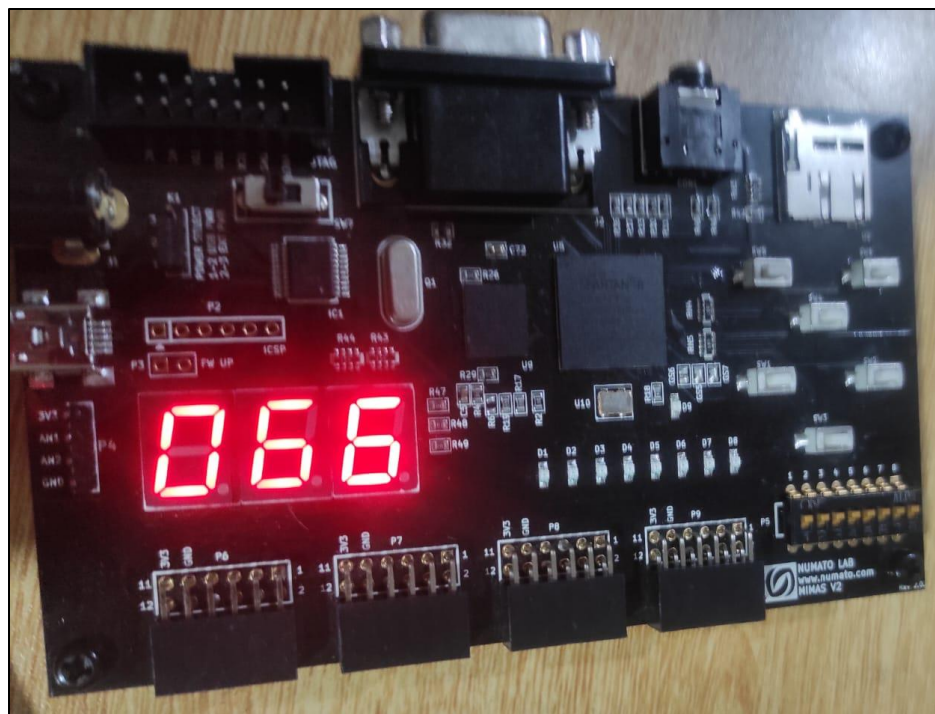


Figure 1.8: BCD Counter Implementation on Spartan6 FPGA Board

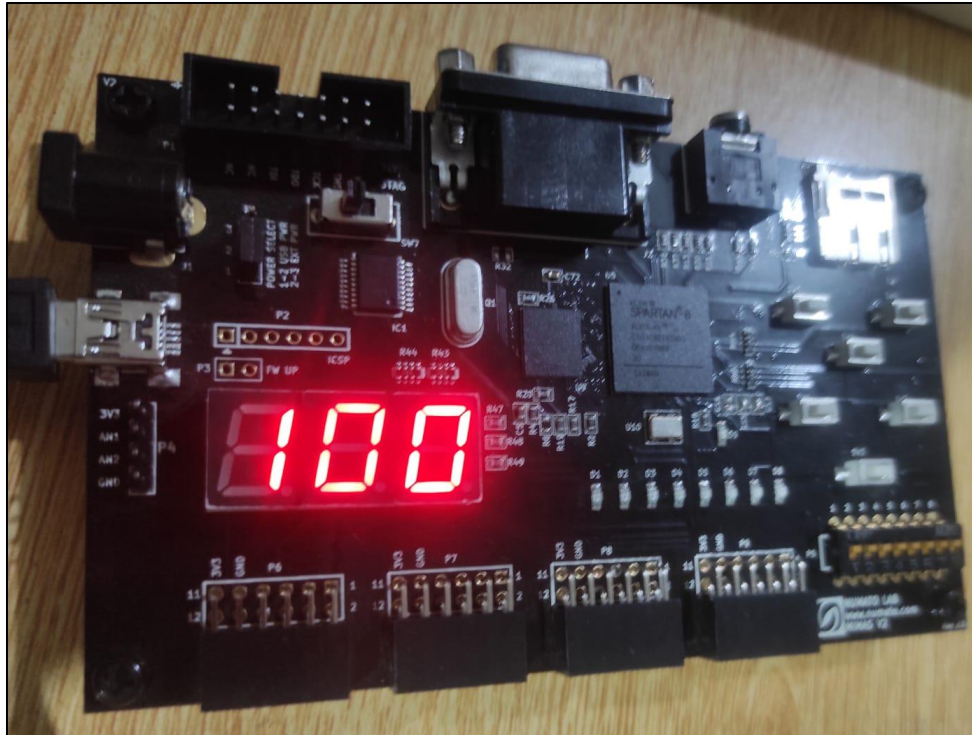


Figure 1.9: BCD Counter Implementation on Spartan6 FPGA Board

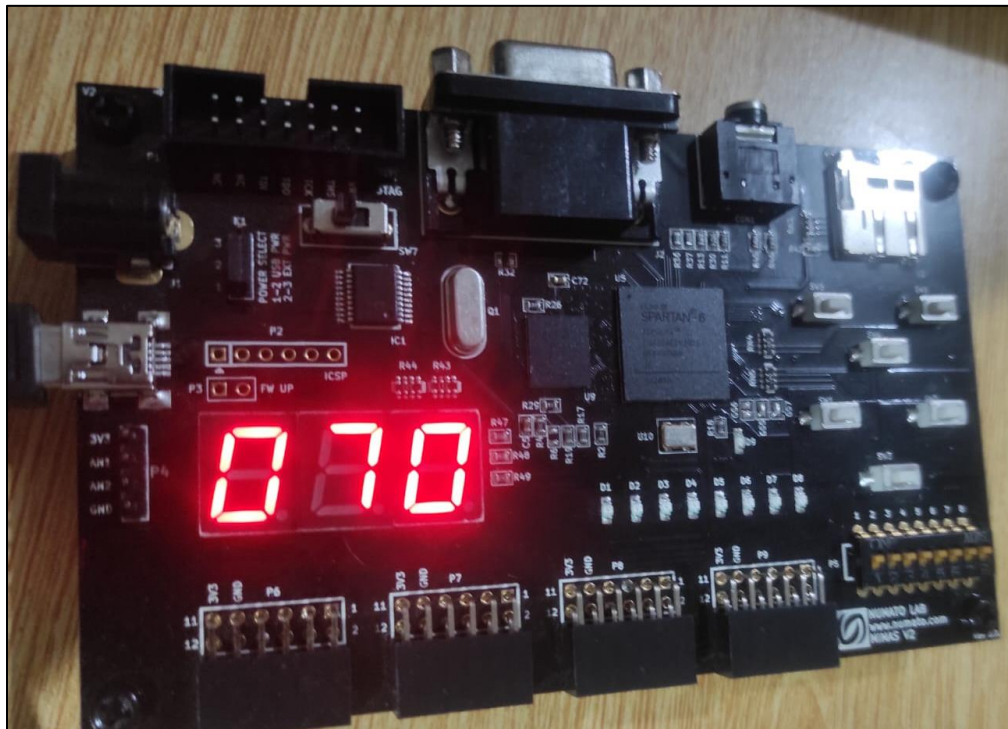


Figure 1.11: BCD Counter Implementation on Spartan6 FPGA Board

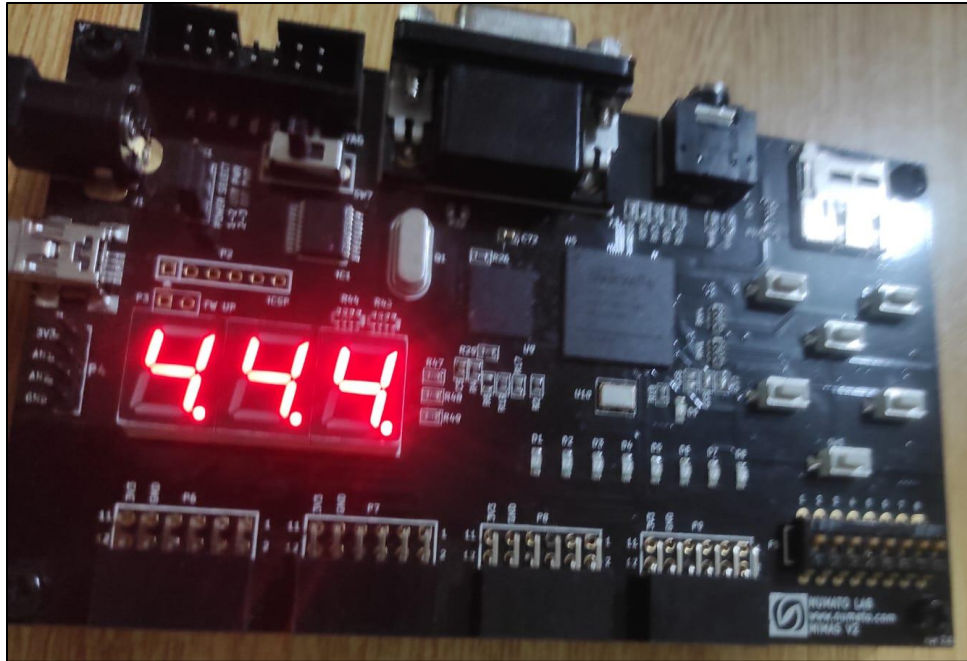


Figure 1.10: BCD Counter Implementation on Spartan6 FPGA Board